

We will start with powers of sine and cosine

Sine & Cosine

Given some integral like $\int \cos^3 x \, dx$ we don't have much to work with, so we will use clever applications of trig identities.

Notice if we let $u = \cos(x)$, $du = -\sin x \, dx$ but $\int \cos^3 x \, dx$ has no "sin x" term. Likewise, we cannot select $u = \sin x$.

Let's fix this. Recall $\cos^2 x + \sin^2 x = 1$

So, $\cos^3 x = \cos x \cos^2 x = \cos x - \cos x \sin^2 x$. So,

$$\int \cos^3 x \, dx = \int \cos x \, dx - \int \cos x \sin^2 x \, dx, \text{ now let } u = \sin x \\ du = \cos x \, dx$$

$$= \sin x - \int u^2 \, du$$

$$= \sin x - \frac{u^3}{3} + C$$

$$\boxed{= \sin x - \frac{1}{3} \sin^3 x + C}$$

Another Example

$$\int \sin^5 x \cos^2 x dx$$

Idea: we want powers of just $\sin(x)$ or just $\cos(x)$ with one extra of the other for du .

with this in mind, substituting $\cos^2 x = 1 - \sin^2 x$ doesn't work as we are left with just $\sin(x)$ terms.

However, since $\sin^5(x)$ is an odd power, we can remove one $\sin(x)$ and write $\sin^4(x) = (1 - \cos^2(x))^2$

then

$$\int \sin^5(x) \cos^2(x) dx = \int \sin(x) (1 - \cos^2(x))^2 \cos^2(x) dx, \quad \begin{array}{l} \text{let } u = \cos x \\ du = -\sin(x) dx \end{array}$$

$$= \int (1 - u^2)^2 u^2 du$$

$$= \int (1 - 2u^2 + u^4) u^2 du$$

$$= \int u^2 - 2u^4 + u^6 du$$

$$= -\left(\frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7}\right) + C$$

$$= -\frac{1}{3} \cos^3 x + \frac{2}{5} \cos^5 x - \frac{1}{7} \cos^7 x + C$$

This Strategy of separating a single $\cos(x)$ or $\sin(x)$ works well for odd powers, but fails for even ones.

We need a new approach for even powers.

Recall

$$\sin^2(x) = \frac{1}{2} (1 - \cos(2x))$$

$$\cos^2(x) = \frac{1}{2} (1 + \cos(2x))$$

Let's use this in an example

Ex $\int \sin^2 x \, dx$

$$= \frac{1}{2} \int (1 - \cos(2x)) \, dx$$

$$= \frac{1}{2} \int dx - \frac{1}{2} \int \cos(2x) \, dx \quad u=2x, \, du=2dx$$

$$= \frac{1}{2} x - \frac{1}{4} \int \cos(u) \, du$$

$$= \frac{1}{2} x - \frac{1}{4} \sin(u) + C$$

$$= \frac{1}{2} x - \frac{1}{4} \sin(2x) + C$$

Ex:

$$\int \sin^4(x) \, dx, \quad \text{use } \sin^2 x = \frac{1}{2} (1 - \cos(2x))$$

$$= \int \left[\frac{1}{2} (1 - \cos(2x)) \right]^2 dx = \frac{1}{4} \int (1 - 2\cos(2x) + \cos^2(2x)) \, dx$$

Since $\cos^2(2x)$ shows up, use $\cos^2(x) = \frac{1}{2} (1 + \cos(2x))$
 $\cos^2(2x) = \frac{1}{2} (1 + \cos(4x))$

$$= \frac{1}{4} \int \left(1 - 2\cos(2x) + \frac{1}{2} + \frac{1}{2}\cos(4x) \right) dx$$

$$= \frac{1}{4} \int \left(\frac{3}{2} - 2\cos(2x) + \frac{1}{2}\cos(4x) \right) dx$$

$$= \frac{1}{4} \left[\int \frac{3}{2} dx - 2 \int \cos(2x) dx + \frac{1}{2} \int \cos(4x) dx \right]$$

$$= \frac{1}{4} \left[\frac{3}{2}x - \sin(2x) + \frac{1}{8}\sin(4x) \right] + C$$

$$= \boxed{\frac{3}{8}x - \frac{1}{4}\sin(2x) + \frac{1}{32}\sin(4x) + C}$$

General way to evaluate $\int \sin^m(x) \cos^n(x) dx$

1) If power of cosine is odd ($n = 2k+1$)

Save one pair of cosine and use

$\cos^2 x = 1 - \sin^2 x$ for the rest and let $u = \sin x$.

$$\int \sin^m x \cos^{2k+1} x dx = \int \sin^m(x) \cos^{2k}(x) \cos(x) dx$$

$$= \int \sin^m(x) (1 - \sin^2(x))^k \cos(x) dx$$

$$= \int u^m (1 - u^2)^k du$$

2) If power of sine is odd ($m=2k+1$),

factor one power of sine, then substitute $\sin^2 x = 1 - \cos^2 x$
and let $u = \cos(x)$

$$\begin{aligned}\int \sin^{2k+1}(x) \cos^n(x) dx &= \int \sin(x) \sin^{2k}(x) \cos^n(x) dx \\ &= \int (1 - \cos^2 x)^k \cos^n(x) \sin(x) dx \\ &= \int (1 - u^2)^k u^n du\end{aligned}$$

3) If both powers of sine and cosine are even, use
half angle identities

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)), \quad \cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

and sometimes another identity

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x)$$

Note: If both powers of sine and cosine are odd,
either method 1 or 2 will work.

Integrals of Secant & Tangent:

Recall $\frac{d}{dx} \tan(x) = \sec^2(x)$ and $\frac{d}{dx} \sec(x) = \sec(x)\tan(x)$.

and $\tan^2(x) = \sec^2(x) - 1$ or $\sec^2(x) = \tan^2(x) + 1$

So, in an integral $\int \tan^m(x) \sec^n(x) dx$

to perform u -substitution we need to factor a $\sec^2(x)$ or a $\sec(x)\tan(x)$.

Ex: $\int \tan^6 x \sec^4 x dx$

$$= \int \tan^4(x) \sec^2(x) \sec^2(x) dx$$

$$= \int \tan^4(x) (\tan^2(x) + 1) \sec^2(x) dx, \quad \text{let } u = \tan(x)$$

$$= \int u^4 (u^2 + 1) du, \quad du = \sec^2(x) dx$$

$$= \int u^6 + u^4 du$$

$$= \frac{1}{7} u^7 + \frac{1}{5} u^5 + C$$

$$= \frac{1}{7} \tan^7(x) + \frac{1}{5} \tan^5(x) + C$$

Ex:

$$\int \tan^5 x \sec^7(x) dx$$

$$= \int \tan^4(x) \sec^6(x) \tan(x) \sec(x) dx$$

$$= \int (\sec^2(x) - 1)^2 \sec^4(x) \tan(x) \sec(x) dx, \quad \text{let } u = \sec(x)$$
$$du = \sec(x) \tan(x) dx$$

$$= \int (u^2-1)^2 u^6 du$$

$$= \int u^{10} - 2u^8 + u^6 du$$

$$= \frac{1}{11} u^{11} - \frac{2}{9} u^9 + \frac{1}{7} u^7 + C$$

$$= \frac{1}{11} \sec^{11}(x) - \frac{2}{9} \sec^9(x) + \frac{1}{7} \sec^7(x) + C$$

In general for $\int \tan^m(x) \sec^n(x) dx$

1) If power of $\sec(x)$ is even, $n=2k$, factor $\sec^2(x)$ out
and use $\sec^2(x) = \tan^2(x) + 1$ and let $u = \tan(x)$

$$\begin{aligned} \int \tan^m(x) \sec^{2k}(x) dx &= \int \tan^m(x) \sec^{2k-2}(x) \sec^2(x) dx \\ &= \int \tan^m(x) (\tan^2(x) + 1)^{k-1} \sec^2(x) dx \\ &= \int u^m (u^2 + 1)^{k-1} du \end{aligned}$$

2) If power of $\tan(x)$ is odd, $m=2k+1$, factor out $\sec x \tan x$
use $\tan^2 x = \sec^2 x - 1$, and let $u = \sec x$

$$\begin{aligned} \int \tan^{2k+1}(x) \sec^n(x) dx &= \int \tan^{2k}(x) \sec^{n-1}(x) \sec(x) \tan(x) dx \\ &= \int (\sec^2(x) - 1)^k \sec^{n-1}(x) \sec(x) \tan(x) dx \\ &= \int (u^2 - 1)^k u^{n-1} du \end{aligned}$$

In other cases, it is not as straightforward, it's helpful to know

$$\int \tan(x) dx = \ln|\sec x| + C, \quad \int \sec x dx = \ln|\sec x + \tan x| + C$$

Ex:

$$\begin{aligned}\int \tan^3(x) dx &= \int \tan^2(x) \tan(x) dx \\&= \int (\sec^2 x - 1) \tan(x) dx \\&= \int \tan(x) \sec^2(x) dx - \int \tan(x) dx \\&= \int \tan(x) \sec^2(x) dx - \ln|\sec x| \\&= \int u du - \ln|\sec x|\end{aligned}$$

$$\boxed{= \frac{1}{2} \tan^2(x) - \ln|\sec x| + C}$$

let $u = \tan x$
 $du = \sec^2 x dx$

If power of tangent is even with odd power of secant
we can express everything in terms of secant and
use integration by parts.

Ex: $\int \sec^3 x dx$, let $u = \sec x$ $dv = \sec^2 x dx$
 $du = \sec x \tan x dx$ $v = \tan x$

$$\begin{aligned}&= \sec x \tan x - \int \tan^2 x \sec x dx \\&= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx \\&= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx\end{aligned}$$

$$2 \int \sec^3(x) dx = \sec x \tan x + \ln |\sec x + \tan x| + C$$

$$\int \sec^3 x dx = \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C$$

Integrals of the form $\int \cot^n(x) \csc^n(x) dx$ are similar
using

$$1 + \cot^2(x) = \csc^2(x) \quad \text{or} \quad \csc^2(x) - 1 = \cot^2(x)$$

Product Identities

To evaluate $\int \sin(nx) \cos(mx) dx$, $\int \sin(nx) \sin(mx) dx$
or $\int \cos(nx) \cos(mx) dx$, use the following

$$a) \sin(A) \cos(B) = \frac{1}{2} (\sin(A-B) + \sin(A+B))$$

$$b) \sin(A) \sin(B) = \frac{1}{2} (\cos(A-B) - \cos(A+B))$$

$$c) \cos(A) \cos(B) = \frac{1}{2} (\cos(A-B) + \cos(A+B))$$

Ex:

$$\int \sin(4x) \cos(5x) dx$$

$$= \frac{1}{2} \int \sin(-x) + \sin(9x) dx = \frac{1}{2} \int \sin(9x) - \sin(x) dx$$

$$= \frac{1}{2} \left(-\frac{1}{9} \cos(9x) + \cos(x) \right) + C$$

$$= -\frac{1}{18} \cos(9x) + \frac{1}{2} \cos(x) + C$$